

Math 526, F06
Quiz 9

Name:
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (12 points) Let $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$, find all eigenvalues and the corresponding eigenvectors of $\mathbf{A}$.

Solution:

$$\begin{aligned} \det(\mathbf{A} - \lambda \mathbf{I}) &= 0 \\ \det \begin{bmatrix} 1 - \lambda & 2 \\ 5 & 4 - \lambda \end{bmatrix} &= 0 \\ (1 - \lambda)(4 - \lambda) - 10 &= 0 \\ \lambda^2 - 5\lambda - 6 &= 0 \\ (\lambda - 6)(\lambda + 1) &= 0 \\ \lambda &= 6, -1 \end{aligned}$$

For the eigenvalue $\lambda = 6$, we solve $(\mathbf{A} - 6\mathbf{I})\mathbf{x} = \mathbf{0}$ to find the corresponding eigenvectors:

$$\begin{aligned} \begin{bmatrix} -5 & 2 \\ 5 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ x_2 &= \frac{5}{2}x_1 \\ \mathbf{x} = \begin{bmatrix} x_1 \\ 5x_1/2 \end{bmatrix} &= c_1 \begin{bmatrix} 1 \\ 5/2 \end{bmatrix} \end{aligned}$$

For the eigenvalue $\lambda = -1$, we solve $(\mathbf{A} + \mathbf{I})\mathbf{x} = \mathbf{0}$ to find the corresponding eigenvectors:

$$\begin{aligned} \begin{bmatrix} 2 & 2 \\ 5 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ x_2 &= -x_1 \\ \mathbf{x} = \begin{bmatrix} x_1 \\ -x_1 \end{bmatrix} &= c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} \end{aligned}$$

Problem 1. (8 points) Let $\mathbf{A} = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & -2 \\ 0 & -2 & 1 \end{bmatrix}$, find all eigenvalues of $\mathbf{A}$.

Solution:

$$\begin{aligned} \det(\mathbf{A} - \lambda \mathbf{I}) &= 0 \\ \det \begin{bmatrix} 1 - \lambda & -1 & 2 \\ 0 & 1 - \lambda & -2 \\ 0 & -2 & 1 - \lambda \end{bmatrix} &= 0 \\ (1 - \lambda)[(1 - \lambda)^2 - 4] &= 0 \\ (1 - \lambda)(\lambda^2 - 2\lambda - 3) &= 0 \\ (1 - \lambda)(\lambda + 1)(\lambda - 3) &= 0 \\ \lambda &= 1, -1, 3 \end{aligned}$$